



What family forest owners talk about when they talk about trees: Bioenergy and forest landscapes in the U.S. South[☆]

John Schelhas^{a,*}, Sarah Hitchner^b, J.Peter Brosius^b

^a Southern Research Station, USDA Forest Service, 320 Green Street, Athens, GA 30602, USA

^b Department of Anthropology, Baldwin Hall Room 250, University of Georgia, Athens, GA 30602, USA

ARTICLE INFO

Keywords:

Family forest owners
Forest landscapes
Bioenergy
Cultural models
Qualitative research

ABSTRACT

Family forest owners in the southern United States have complex relationships with trees and forests rooted in a long history of forest-based rural economies and land use change. As the South is the primary wood-producing region in the U.S., economic and utilitarian valuations of forests are stronger here than in other parts of the country. However, forests are also deeply valued for aesthetics, recreation, wildlife (game and non-game), family history, and community heritage. Social and cultural values of forests are intertwined with economic values and biophysical conditions, but family forest owner decision-making is still not well-understood. The development of bioenergy in the South provides an opportunity to examine the social and cultural aspects of trees and forests and their role in decision-making. We analyze the ways that family forest owners and foresters talk about trees and forests in interviews and field notes from ethnographic fieldwork on wood-based bioenergy in two communities in Georgia and one in Mississippi to understand the multiple, complex, and sometimes competing values ascribed to southern forests, forest landscapes, and particular tree species. Bioenergy development serves as both a driver of new markets for wood products and as a lens through which to examine how owners make decisions about how to best manage their forests for multiple objectives.

1. Introduction

Forests in the U.S. South are part of complex socio-ecological landscapes in which forest characteristics have long reflected human activities. People have shaped and lived within these landscapes for thousands of years in ways that reflect interwoven materialities, values, and social relationships (Cowdrey 1996; Fuentes 2019). These landscapes—valued for diverse forest products, environmental services, and cultural reasons—have changed over time and continue to change. Family forest owners have become an important element of these landscapes over the past century (Butler and Wear 2013). The forest industry and government programs for landowners have also shaped the current landscape of the U.S. South, which remains heavily forested. Family forest owners operate within these existing landscapes, while at the same time shaping future landscapes through their decisions.

Family forest ownership and decision-making has been the subject of considerable research, yet family forest owner behavior remains poorly understood (Butler et al., 2023). We have long known that their harvest

decisions differ from those of corporate owners, which generally conform to economic models (Butler and Sass 2023; Newman and Wear 1993). In general, family forest owners' management and harvest decisions respond to various biophysical, social, and economic factors, as well as incentive programs and information, but they are difficult to model because they are heavily influenced by social and cultural factors (Butler et al. 2023; Butler and Sass 2023). Qualitative studies can add to our understanding (Butler et al. 2023). A variety of concepts, such as integrated landscape approaches and socio-ecological niche theory, focus on social and cultural aspects of forest use and management as deeply entangled with biophysical and economic factors in shaping landscapes (Gebru et al. 2019; Ojiem et al. 2006; Sayer and Boedhihartono 2018).

Southern forests are one of the nation's and world's largest sources of roundwood, pulp, and, increasingly, wood for bioenergy (Brandeis et al. 2022, Mei 2019). While slightly less than one-third of the South's forests are in corporate ownership (often timber companies and timber management entities), about half of all forests in the South are owned by

[☆] This article is part of a special issue entitled: "Family Forest Landowners of the United States: Exploring Challenges and Opportunities in Light of the Sustainable Forestry Paradigm" published at the journal *Trees, Forests and People*.

* Corresponding author.

E-mail address: john.schelhas@usda.gov (J. Schelhas).

<https://doi.org/10.1016/j.tfp.2024.100606>

families (USDA Forest Service 2021a, b). Reasons for owning family forests in the U.S. Southeast are complex, but in spite of the importance of forest product industries in the region, timber is only one of many motivations for owning family forest. The National Woodland Owners Survey (NWOS) finds that, across the South, family forest owners value objectives such as beauty or scenery, nature protection, wildlife habitat, hunting, recreation, family legacy, and investment over timber production (USDA Forest Service 2021a, b). Approximately one-quarter of forest owners have cut trees for sale in the past five years (USDA Forest Service 2021a, b), indicating the importance of timber.

This paper adds to previous studies by examining family forest owners' complex relationships with forests and forest management through cultural models drawn from transcripts of conversations with family forest owners and forestry professionals. Here we show how the introduction of wood-based bioenergy facilities into rural communities in the U.S. South with long histories of engagement with forests and forest products illuminates the various ways that family forest owners think about forested landscapes in general and specific tree species within them. Wood-based bioenergy will change forest landscapes, but outcomes will be determined by family forest owners' decisions that are rooted in social and cultural relationships to forests as much as they are in markets and economics.

2. Background

Forests, and people's relationships with forests, are complex and everchanging yet always rooted in history. A number of different Native American groups altered the southern environment, primarily with fire over at least the past 12,000 years, creating the mosaic of open and uneven-aged forests, native settlements, agricultural lands, and prairies encountered by early European settlers (Carroll et al. 2002). Colonial settlers used fire to clear land for homes and farms, and forests for timber and naval stores (Carroll et al. 2002; MacCleery 2011; Owen 2002). In the 20th century, intensification and geographic shifts in agriculture, along with industrialization and urbanization, led to increasing forest cover through forest regeneration and tree planting (Rudel 2023; Williams 1989). Present-day forests have been shaped by the agricultural intensification and in some places, decline of agriculture leading to natural regeneration; by demand for pulpwood and timber by the wood products industry; and by pine planting by industry and family forest owners—the latter often under government assistance programs (Rudel 2023; Schelhas et al. 2021). Forests will continue to change in response to changes in the forest product industries, government policies, and the diverse forest values of landowners and society.

In the early 2000s, a significant interest in bioenergy began to develop in the U.S. and Europe. Bioenergy was seen as the best option for achieving a renewable energy future in terms of environmental, social, and political sustainability (Brown 2012). Bioenergy using woody biomass was supported by policies in both the European Union and United States (Aguilar 2014; Dwivedi and Alavalapati 2009; Lantinen et al. 2014, US DOE 2006), and additional incentives in agriculture, rural development, and forest sectors also supported these goals (Lantinen et al., 2014). Interest in bioenergy surged in the early 2000s in response to rising fuel prices; this interest was supported by significant levels of government funding and promotion leading to a near-frenzy of bioenergy start-ups and research that lasted into the mid-2010s (Hitchner et al., 2022). Liquid fuels from cellulosic biomass were the primary goal, although pellet production as a renewable¹ source for power generation also surged. In spite of considerable government and private investment, the path to economically competitive liquid fuels from trees was difficult. Many of the most hyped efforts failed, although those with a more methodical approach have continued to make

progress (Hitchner et al. 2022). More recently, any rise in gas prices may revives talk about bioenergy for liquid fuels, although most rises have been temporary and the future of liquid fuels from woody biomass is uncertain. On the other hand, the manufacturing process for wood pellets was already known, and exports of pellets to Europe to replace coal for power generation have continued.

The U.S. South is the most forested region of the country, and also has the most forests in private ownership, both family and corporate (Nelson et al. 2020; USDA Forest Service 2021a, b). Family forest owners supply more than 40 % of wood products in the U.S. and are important to bioenergy development (Butler and Sass 2023). This is especially true in the U.S. South, where wood-based bioenergy projects have been initiated due to the abundance of available woody biomass. Because so much land is privately owned in the South, forest owner involvement in supplying these bioenergy facilities is seen as integral to the successful implementation of wood-based bioenergy.

To date, much of the woody biomass used in the bioenergy industry has been pulpwood for wood pellets made available by declines in the pulp and paper industry (Brandeis et al. 2021, Schelhas et al. 2018). In terms of purchases of woody biomass for both liquid fuel and pellet plants, proposed plants nearly always told communities that they would use waste wood, such as tops and limbs, but the costs of gathering waste wood along with the processing issues caused by dirty waste wood resulted in boles from pulpwood-sized pine trees becoming the sole source material at our research sites (Schelhas et al. 2018). But bioenergy research has also focused on new species, including hardwoods and eucalyptus, and silvicultural changes, such as shorter rotations, changing planting densities, and new mixes of genetic material (Perdue et al., 2017, Stanturf et al. 2013). How family forest owners have reacted, and will continue to react, to these proposed changes in environmental policies, energy demands, land use configurations, species composition, harvesting practices, and end uses for wood products will drive both the future of wood-based bioenergy production and forest landscape change in the U.S. South.

One factor in the choice of location for bioenergy facilities is the availability of woody biomass surrounding the proposed plant location. It is often assumed that forest owners seek to maximize economic returns from forests, and researchers often experiment with new species, planting systems, and wood products under this assumption (Hitchner et al. 2022). But ignoring social and cultural factors risks misunderstanding family forest owners' motivations and choices. It cannot be taken as a given that forest owners will be willing to change their existing practices for new management practices or end uses. Family forest landowner willingness to harvest biomass has been found to be price sensitive in economic studies, although saw timber prices generally drive harvest decisions (Aguilar et al. 2014a, 2014b). Paula et al. (2011) also found price and size of timberland holding to be key factors in willingness to supply biomass, although they noted interest in the potential of bioenergy from local feedstocks to promote local economic development and address climate change. Joshi and Mehmood (2011) found that younger landowners with larger landholdings were most likely to supply woody biomass for bioenergy. Policy supports and tax incentives can increase economic viability and participation in biomass markets (Adjoyi and Kebede 2017, Bailey et al. 2011). Social studies have found that family forest landowners' willingness to provide biomass was linked to their motivations for owning land, reflecting differences in economic and conservation orientations (Aguilar et al. 2013). These papers establish a general interest among family forest owners in producing woody biomass, albeit contingent on favorable economic factors and varying with demographics, forest size, and forest values. These studies were carried out though general forest owner surveys that were not geographically focused on specific bioenergy developments, and most took place amid the early enthusiasm for bioenergy development. Further studies on the social and cultural factors that shape family forest owners' decisions and under concrete conditions of operation and markets under bioenergy plants can further

¹ Renewability and how it is determined has often been questioned, see Hitchner et al. 2002 and Hitchner et al. (this issue).

develop our understandings of both family forest owners and bioenergy development.

In this paper, we aim to take a closer look at how family forest owners and forestry professionals talk about trees, and then examine this talk to develop a more nuanced understanding of their thoughts, values, and practices. We do not examine family forest owners in isolation or as individual agents, but rather as actors within the complex social networks of forestry in which they are embedded (Hitchner et al. 2019). Forest landowners and forestry professionals (including county extension agents, county forestry committees, employees of state forestry agencies and federal forestry agencies, and forestry non-profits) frequently come together at conferences, workshops, and forest land field days to discuss forest management and the changing forest product environment. It is that at events like this that shared cultural models are formed and reinforced. Our research takes place through in-depth ethnographic research, rather than surveys, which includes participation in community forestry events and provides a more nuanced understanding of landowner value, motivations, and behaviors. Our research was conducted as bioenergy development was being implemented and often struggling, and was focused in the areas around bioenergy plants constructed and operating at least at a pilot level.

3. Theory, methods, and study sites

The way forest owners think about forests and trees, in combination with biogeography and forest markets (among other things), shape forest landscapes. New and different forests arise, but they have their roots in forest history and emerge from existing forest landscapes and forest owner values. Talking to people about their forests and potential changes reveals the nature of current socio-ecological systems and provides insight into future changes.

3.1. Theory

In this paper, we draw on our research on cultural models associated with trees to investigate how, specifically, forests might change in the future as bioenergy development continues. The cultural understanding of trees and forests can and ultimately does shape forests themselves (Schelhas and Pfeffer 2005, 2009). Cultural understandings can be teased out of discourse (Quinn 2005), with analysis based in a methods from cognitive anthropology, including cultural models (D'Andrade and Strauss 1992, Holland and Quinn 1987, Strauss and Quinn 1997). Discourse—or, more simply, talk—defined as spoken or written language in use, can be gathered from interviews for the empirical study of beliefs, patterns of thought, and values—which can then be linked to motivation and behaviors (Quinn 2005a, Schelhas and Pfeffer 2005, Strauss 1992).

People in social groups share understandings² that have been learned and internalized through shared experiences, and use such shared understandings to make sense of and organize the ways they operate in the world through their thoughts, feelings, motivations, and actions (Quinn 2005a, 2–3). Such cultural ideas and assumptions may be internalized to an extent that they are taken for granted by individuals rather than being explicitly considered (Strauss 2005). Cultural analysis of discourse provides the best window we have into cultural understandings and their use by individuals because these understandings are often implicit and must be teased out of interview data (Quinn 2005a). Cultural models, or schemas, are represented by key words, phrases, metaphors, patterns of reasoning, and other shared language that provide windows into shared understandings (Holland and Quinn 1987, Quinn 2005b). Cultural models reveal motives and identify cultural complexities (D'Andrade and Strauss 1992, Strauss 2005), thus providing both general cultural understandings and identifying point of divergence. While

motives identified from discourse do not necessarily predict behavior, analysis of cultural models within ethnographic research provides additional empirical insights that guide and support discourse analysis. Individuals are social and cultural actors, and narratives and cultural models are co-produced within social and environmental networks (Lejano et al. 2013).

3.2. Methods

Using a complementary array of qualitative social science methods (Bernard 2011), we conducted ethnographic research in three communities in Georgia and Mississippi with different types of bioenergy facilities. We spent three months living in each of these three main field sites, attending meetings, workshops, and events and conducting interviews. We interviewed many different stakeholders, including a wide array of forest owners, community members, local politicians, cooperative extension agents, and loggers and others employed in the forest industry. For these interviews, we employed a mix of structured and semi-structured interview methods. Interviews were conversational in order to give narrative advantage to the interviewee and capture the way people normally talk to the extent possible. Within these conversations, we discussed preidentified topics that included: time and role in the community, the local forest landscape, thoughts and opinions about bioenergy, sustainability and certification, the role of government in bioenergy and forestry, the local bioenergy facility, forest ownership and management, and how bioenergy might impact their own forest management (see supplement). During interviews, we took detailed notes on both questions and responses and immediately transcribed them. We also transcribed fieldnotes about the location of the interview, relevant observations about the interviewee, and our reflections on the interview. We sought to capture diversity representative of the forestry community in the interviewee pool, and also asked interviewees to recommend other potential interviewees including those who might think differently than they did. We conducted a total of about 175 interviews with forest owners, forestry professionals, and community members (with many interviewees being two or more of these; interviews lasted between thirty minutes and three hours (averaging about an hour).

We used NVivo qualitative analysis software to conduct content analysis of ethnographic data collected in our three primary field sites. NVivo is a qualitative software program used to analyze unstructured text, audio, video, and image data from various sources through thematic coding and search for patterns among data within those codes. Using NVivo, we read through the transcripts and data to identify key themes and patterns, and then used this to develop a coding system that was applied to the data. As such, codes reflect key words and phrases, shared thought patterns and metaphors used by our respondents. We identified key themes related to these cultural models, which form the framework for the data section of this paper. We discuss cultural models within each theme, and include representative quotes to illustrate them. Our goal was to identify shared cultural models and patterns from our interviews, rather than compare and contrast individuals. We used snowball sampling to identify interviewees during the process of ethnographic research; given this sampling method, comparison by demographic characteristics and forestry roles would only be misleading, and is thus not reported.

3.3. Study sites

Our place-based ethnographic research was conducted from 2012 to 2015 in three primary field sites representing differing suites of social and ecological factors: Soperton, Georgia; Columbus, Mississippi; and Waycross, Georgia. Although they differ somewhat in biogeography, each was the site of a bioenergy plant using new technology with outside investment and government subsidies that stimulated both hope and suspicion about long-term viability (Hitchner et al. 2016, 2017, 2022).

² Social groups have loose boundaries and may overlap, and understandings may be shared to greater or lesser degrees.

3.3.1. Soperton, Georgia (*Range fuels/lanzatech*)

Known as the “Million Pines City,” Soperton has a long history of modern forestry, including some of the earliest pine plantations and pulp production. The community leaders embraced the construction of a cellulosic ethanol plant, Range Fuels, in November 2007, in hopes of being at the forefront of a new forest product. Range Fuels was expected to produce 40 million gallons per year of cellulosic ethanol using gasification technology and southern yellow pine³ as a feedstock but only produced one batch of methanol. Following bankruptcy, LanzaTech purchased the facility at auction for \$5.1 million, renamed it the Freedom Pines Biorefinery, and has focused on wood-based aviation fuel.

3.3.2. Columbus, Mississippi (*KiOR*)

A small industrial city in east central Mississippi lies at the confluence of pine forests, flood plain hardwoods, and Black Belt prairies. In 2010, KiOR built the world's first commercial-scale cellulosic biocrude plant in Columbus, Mississippi, which began production in 2012. It used a proprietary biomass fluid catalytic cracking (BFCC) technique to convert biomass feedstock, specifically southern yellow pine, into crude oil that could be refined into gasoline, diesel, and aviation fuels. The plant was the first to produce a commercial liquid fuel from wood. After our research was completed, KiOR was unable to produce an economically competitive fuel and filed for bankruptcy.

3.3.3. Waycross, Georgia (*Georgia biomass*)

Located in coastal plain pine forests with strong wood products industries, Georgia Biomass began operation in 2011 with the capacity to produce 750,000 tons of pellets per year from local forests—requiring about 1.5 million metric tons of fresh wood per year. Pellets, unlike cellulosic liquid fuels, were a proven technology and we included a pellet plant to gain insight into community and forest owner perspectives to actual harvesting of wood for bioenergy. Georgia Biomass is a wholly-owned subsidiary of the German utility company RWE Innogy, and these pellets are shipped from the port in Savannah, Georgia, to supply biomass power plants and co-generation facilities in Europe.

4. Results

4.1. Forest values

As we noted above, family forest owners, when surveyed, indicate that they have many reasons for owning forest land that are more important than timber production. Yet, timber is important to many forest owners, sometimes as an important economic activity in itself and sometimes as a financial means to support forest ownership for other values. Within forestry circles, forestland is seen as a good investment and timber harvest is seen as of primary importance. For example, a forester in Mississippi saw forests as a “cash cow” and forest product markets as important for maintaining forest ownership and health. Yet forest owners balance timber with many other objectives. Several forest owners in Soperton, GA expressed this clearly. One said, “When I first owned timberland, it was primarily to make a profit. The older I got, the more I cared about the aesthetics.” Forest owners expressed other very specific values of forests, such as recreation hunting, beauty, and investment. The complex interaction between timber and other values is evident in an industry forester's statement; he said, “I don't like to see 100 % pine trees. I like healthy forests. It is not healthy if it is only trees. I'm a forester, but I like to see quail in a forest. It all has its place.”

There is evidence that forest owners value the forest landscape itself, which includes a forest products industry. Many noted how the presence of forests adds value to the landscape beyond their economic

importance. Citing the responsibility of forest stewardship, one forest owner said, “I am a true conservative. I love trees, but I don't hug trees. I'm very conservative. I think we should leave the planet better than we found it.” This love of the landscape can be seen in the statement of another forest owner, while standing on the porch of his retirement house surrounded by pine plantations: “I love having forest around me. It's sort of like a blanket. I can be next to God, go outside, walk the trails through the forest. Feel Mother Nature, see wildlife in the forest, the whole gamut. The forest is my blanket.” Forest owners in the U.S. South do not necessarily distinguish between natural forest and pine plantations, saying they are accustomed to timber harvesting and logging trucks—as one forest owner said, “unlike urban environmentalists.” Rather, although they live in a forested landscape where wood product markets shape forest composition, those forests have deeper cultural and personal meanings to them.

4.2. The changing landscape

Many family forest owners have a long history living in their forest landscape, and they told stories of earlier times from their lives and those of their ancestors. Many of these stories go back to the turpentine era, which was remembered with both fondness and consternation. The turpentine, or naval stores, industry provided an important and profitable land use option for forest owners prior to the late 20th century. Many forest owning families had been involved in the turpentine business and considered it to have been very profitable. One community member said, “It's a rural county, and was all farm and turpentine.” A forest owner said, “I wish turpentine was still going. My dad made a lot of money from turpentine.” These statements were almost always followed by noting that the turpentine business had faded out in the 1970s, and that most turpentine now comes from China. As one forest owner noted:

I grew up hearing about pines. Forestry was my 4-H project. My husband worked in the gum naval stores rolling barrels and running the dip wagon and marking trees. Turpentine was a huge industry, and people made money in it. It was very labor intensive and physically demanding. If you did it, the money was good, but you had to be strong. It has died out now because of foreign competition from China. It's been dying out for the last 20 years. [We] were one of the last to still do turpentine.

The roots of the pine forest landscape for current community members lie in the turpentine era, as turpentine led to tree planting. As one forest owner noted, “James Fowler planted pine trees for the turpentine business. Before that, everything was natural regeneration. ... He started the forest growing industry.” The pulp and paper industry, along with sawtimber, soon followed. Soperton, for example, became the “Million Pines City,” with the *Soperton News* being the first newspaper printed on paper made from pines when Charles H. Herty devised a process to make quality paper from pine trees. There were numerous sawmills cutting pine for paper and logs. Most pines were slash and longleaf, although timber companies planted loblolly in 20-year rotations for paper and some forest owners did, too. Over time, farming declined and was replaced by pine trees planted through government programs to such an extent that some people thought that it caused an oversupply of pine trees that led to low wood prices.

Loblolly has often been planted because of its rapid growth, particularly for paper mills. However, as the paper industry declined, government programs have favored longleaf for biodiversity, disease, and fire resistance. Many people rely on timber for retirement income, and harvesting is considered essential to forest health in planted pines. But forests have cultural value, too. One forest owner said, “The forest is in your blood,” and another said, “People have an emotional attachment to land. They keep it in the family for generations.” As one forester said, “[People] want to leave a legacy, leave it in better shape than they found it. [They] are also interested in timber and wildlife.” Forests became a

³ Southern yellow pine is a generic industry term for all the pine species in the southern United States.

part of life in these communities, and bioenergy development was seen as compatible with those values and as a bridge linking the communities' historical dependence on forest products with new end uses for native tree species.

4.3. Hardwoods versus softwoods

We have noted that the general socio-ecological forest landscape is of value to people, and that this landscape includes the forest product industry—sawmills, pulp and paper mill, logging crews and trucks, and forest plantations. The wood products industry, primarily based on southern pine trees, shape the way forest owners view and value individual tree species in the landscape. Several foresters reflected on the importance of pine trees in the region. One said, "I would not plant any hardwood when we are surrounded by pines." Another put this in historical context, noting, "Pine trees are always so great. You think they're maxed, then they find something new ... turpentine, saw timber, paper." Pine trees are further encouraged by government incentive programs. As one forest owner in Mississippi noted, "The CRP [Conservation Reserve Program] has been a gravy train to me. ... I got sucked into the pine tree deal," while noting that there have been diverse markets for hardwoods as well, such as pulp, timber, and cross ties. In the area around Columbus, Mississippi, which is a mix of hardwood and pines, hardwoods were often seen as worth maintaining. One forest owner said, "Hardwoods are backup. When things are slow, I can sell them. There's always demand for good hardwood."

Aesthetic and wildlife-oriented forest owners value hardwoods as well. At a Soperton garden club meeting, one attendee said, "I love the hardwoods so much more than the pines. I really like the ... Cherokee dogwoods and the redbuds.... They shouldn't destroy the big oaks in the woods. They should go around them, be more selective when they do the logging." A forest owner in Mississippi who hunted said, "This here [cherry tree] is a pie tree. The big boys [bucks] come for that pie." Another Mississippi forest owner said, "I have two 100-acre plots. They're pine and hardwoods. I took out the trash trees – gum, hackberry, ash. Anything that wasn't bearing. I left the native hardwoods there. Now I plant a mixture." Another forest owner noted how a timber company had killed all the hardwoods to grow pine, saying "Do you see anything there for a deer to eat? Then you want to lease it for hunting and get high dollar?" Hunting leases were mentioned as bringing in sufficient funds to pay taxes on the land, but forest owners who did not lease out hunting rights also mentioned the role of hardwoods in supporting wildlife. Forests are shaped by what people value in them at a particular place, including both hardwoods and softwoods. Our Mississippi site offered slightly more opportunities for hardwood, both for forest products and hunting, and hardwoods were more valued there.

4.4. Pine species

Although southern yellow pine trees dominate most of our research sites, forest owners differentiate between different types of pine species. Loblolly pine (*Pinus taeda*), slash pine (*Pinus elliottii*), and longleaf pine (*Pinus palustris*) are valued for different reasons and in different social and ecological contexts. Loblolly pine has often been favored in the forest product industry and some government incentive programs because of its rapid growth and short time to harvest. There is more loblolly than any other planted pine species in the South (Zhou et al. 2023). Yet other pine species were more valued by many forest owners. One forest owner who was planting trees in Soperton under the CRP program said, "I've got some loblolly, but I won't ever plant it again." Another said:

When we convert from crop to pines, we're putting in all slash. The advisors on the timber convinced me that loblolly's better I'm not sure that's the case. [In] hindsight I would have planted all slash and no loblolly.... I like longleaf the best. But it requires so much growing

time.... the loblolly won't be replaced. I'll put in more longleaf. I'll put it in for my grandchildren.

There are clearly diverse values of trees that lead to preference for certain trees as well as mixtures of species. One forest owner in Waycross said, "Loblolly makes the best pulp, not as strong boards. Slash has more versatility." Often, this preference for mixed species across the region reflects forest owners' understanding of how species diversification can serve a variety of values and markets.

However, some forest owners have clear preferences for certain species on their own land, usually longleaf pine if their land is suitable for it. One forest owner in Waycross said, "I've got 150 acres of longleaf.... The straw on the longleaf is worth more than the tree." Many forest owners in Georgia mentioned that they had planted longleaf for pine straw⁴ because the market is strong and you could make good returns (see Dickens et al. 2012). Many mentioned that you could not rake pine when you planted trees under the Conservation Reserve Program (CRP) but suggested only staying in the program for ten years and then raking pine straw for the next twenty.⁵

Loblolly as seen a primarily a pulp or bioenergy tree. One experienced forester in Soperton said, "There's not much loblolly here. Some paper companies and private forest owners plant it because it grows faster. It's good sawtimber, but slash is better." A sawmill manager disagreed about the timber value:

Loblolly makes pieces of shit. They're ugly. I don't like them. The limbs are all in your face when you're mowing underneath them. Slash bark is prettier. They're about the same height, but the [diameter] is bigger in slash. I like longleaf better. But that's not a viable option for rotation. It takes too long.

Longleaf was seen as being more expensive to plant, but more resistant to diseases, insects, and fire. There were differing opinions about whether it grew too slowly to be a viable timber tree, with many forest owners thinking it would only be harvested by their grandchildren. Others noting that new planting material and techniques had shortened rotations. The Columbus, Mississippi region was too far north for longleaf pine, and people there were happy with loblolly as a pine although many preferred hardwoods. In sum, the full range of suitable trees, market opportunities, government programs, and aesthetic values determine tree choice within the forest landscape.

4.5. Intensive management and new trees for bioenergy

Within the forest industry, forests are often managed for maximum production of desired forest products from a single type of tree. Thus, it is no surprise that, when the bioenergy industry began to develop in the South, foresters talked about fast-growing plantations of native pine and non-native eucalyptus trees to supply the bioenergy plants. In Soperton, the Range Fuels company established a small plantation of GMO frost-resistant eucalyptus trees as a demonstration of a potential input for their plant. Forest owners, however, were often wary of new trees planted for a single market, as they are familiar with a variety of introductions in the South—such as paulownia trees and Asian pears—where the markets for the promoted new trees never materialized. Furthermore, with native pine plantations, the early thinnings of small diameter trees for pulp or bioenergy barely pay for themselves; the real return is in the sawtimber at the end of the cycle. Even with sawtimber, many forest owners endeavor to time harvests to favorable sawtimber markets. Switching to all short rotation, small diameter trees to feed bioenergy plants is of little interest for forest owners from an economic

⁴ Longleaf pine needles retain their rich brown color when used in landscaping, whereas loblolly pine needles turn grey.

⁵ Pine straw cannot be raked when land is enrolled in the CRP. Initial CRP contracts are generally for ten year, with an option to renew.

standpoint. One forest owner said, “You cannot pay for a place growing pulpwood.” Another forest owner said, “I’m all for [bioenergy]. But I wouldn’t grow just for that. I don’t think it will ever be an industry in itself. People need to have multiple products, ... you can’t put all your eggs in one basket.” Additionally, as we have shown, family forest owners highly value the non-economic benefits of forests, and dense, fast-growing plantations of native or exotic trees are unlikely to fulfill that interest.

5. Discussion and conclusion

Our research identifies general cultural models, as well as differences among them, through qualitative analysis of transcripts and field notes from interviews with family forest owners and forestry professionals from areas around three bioenergy plants in the U.S. South. Taking a view of cultural models as a generic version of some aspect of the world that is shared to a greater or lesser extent (Quinn 2005b), we have identified and described cultural models that occur in five topical areas relevant to family forest owners and bioenergy. We found that forest values are complex, and that forest ownership is often less an economic endeavor than a means to support forest ownership for social and cultural values. Forests are not generic and interchangeable, but rather rooted in a historic and ever-changing socio-ecological landscape. Preferences among hardwoods and softwoods varies primarily in relation to region and by importance of wildlife. Among softwoods, despite the widespread encouragement of the use of loblolly pine in tree plantations, slash and loblolly pines were often preferred. The multiple values of forests, the deep experience in the historical forest landscape, and dissatisfaction with fast-growing species like loblolly pine create a resistance to research- and extension-developed plantation systems involving short-rotations and fast-growing exotic species. While questionnaire research on forest ownership motivations and willingness-to-harvest is important, our research provides a level of detail and nuance that can only be obtained from in depth qualitative and ethnographic research. By embedding family forest owners within their forestry networks, and considering decision-making within the scope of forest history and change, our research highlights the ways that family forest owners operate in larger contexts rather than as independent actors operating only in their economic self-interest.

Our results show how trees and forests in the South are components of complex socio-ecological landscapes. We have long known that family forest owners value their forests for many reasons other than timber, but talking with forest owners reveals how they balance timber with values such as beauty or scenery, nature protection, wildlife habitat, hunting, recreation, family legacy, and investment. These diverse values are widely shared, but the balancing is done by individual forest owners based on the particular suite of values that is important to them, by market conditions, and the biophysical properties of their land and region. Thus, decision-making will likely always seem somewhat idiosyncratic, although general patterns will emerge. People often talk about timber, pulp, and biomass when they talk about their forests, and it is clear that economic returns are a fundamental part of forest ownership for them. Very few of the forest owners we talked to plan leave their forests untouched. But forests and the trees within them are not viewed as future timber sales, either. They value wildlife, but do not think of their forests as nature preserves. Economic returns from timber might often be better thought of as enabling forest ownership, rather than being the goal of it. People value the aesthetics of certain pine species, as well as hardwood trees. People have an identity as forest owners and want to leave a legacy of a healthy and productive forest.

The overall forest landscape around the communities we worked in is highly valued, and this landscape includes pine plantations and timber harvesting. Forest owners in this region expect to see forestry operations and distinguish themselves from urban environmentalists in that regard. They feel that robust forest products markets serve as a way to “keep forests in forests” and to ensure that future generations will continue to

see the value in forestland ownership.

But landscapes are rooted in the past, and people recall that past when they talk about them. To put it succinctly, landscapes are memory. This is not to say they never change or cannot be changed, but that while people adapt to and understand change, certain elements—for example, pine trees or specific pine species—provide a meaningful bridge across change over time.

Various options that have been explored in research and promoted in extension programs do not fit well within the way forest owners view their forests. Loblolly pine in general is viewed much less favorably than slash or longleaf, and hardwoods are more appreciated more north of the areas where slash and longleaf can be grown. Short rotation pine plantations are unattractive both for aesthetic and economic reasons, with sawtimber being seen as economic engine of forestry. Fast growing exotic species do not fit within historic forest landscapes, and people are very wary of new trees with for a single, speculative use. Trust in, as well as appreciation for, pines has been built through landscape and economic changes over more than a century. Forests, even tree plantations, are a long-term endeavor, and change in forest management and harvesting is often slow. This is not simply a resistance to the new, as bioenergy products that fit within their landscapes and values were accepted, particularly in the way they provided a new market for pulp wood.

Research, technology development, and extension will all be improved by considering the ways that family forest owners think about and value their trees and forests. Cultural analysis of how people talk about tree and forests, and placing these ways of talking about them in historical, social, cultural, economic, and ecological context, leads to a more nuanced understanding of family forest owner values and choices. This is vital as new wood-based bioenergy continues to develop here, because technological advances will likely overcome the obstacles that have hindered the success of specific bioenergy projects in the past (such as with LanzaTech and KiOR). Technology will almost certainly create new uses for native pines and hardwoods (and genetically modified versions of these that are less integrated into the natural ecosystems), as well as for introduced exotic species. Forested landscapes will continue to change, as future generations take over the current management and impose new perspectives and objectives and as new markets for natural resources and new land use opportunities emerge. Foresters and agencies tasked with assisting family forest owners, as well as those in power to make decisions about bringing in new bioenergy facilities to rural communities in the South, need to be cognizant of the multiplicity of forest owner values and other factors that influence their decision-making processes. The first step is in listening closely to how family forest owners talk about trees.

Funding

This work was supported by USDA NIFA-AFRI (National Institute of Food and Agriculture–Agriculture and Food Research Initiative) Sustainable Bioenergy Challenge Area (USDA-NIFA Award 2012–67009–19711).

CRedit authorship contribution statement

John Schelhas: Writing – original draft, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Sarah Hitchner:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **J. Peter Brosius:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

John Schelhas, reports financial support was provided by US Department of Agriculture. John Schelhas reports a relationship with US Department of Agriculture that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.tfp.2024.100606](https://doi.org/10.1016/j.tfp.2024.100606).

References

- Adjoyi, G., Kebede, E., 2017. Landowners' willingness to supply woody biomass for biofuel in West Alabama. *Profess. Agricult. Workers J.* 5 (1), 4.
- Aguilar, F.X., 2014. Wood energy in the E.U. and U.S.: assessment and outlook to 2030. In: Aguilar, F.X. (Ed.), *Wood energy in developed economies: resource management, economics, and policy*. Earthscan, London, pp. 306–327.
- Aguilar, F.X., Daniel, M.J., Narine, L.L., 2013. Opportunities and challenges to the supply of woody biomass for energy from missouri nonindustrial privately owned forestlands. *J. For.* 111 (4), 249–260.
- Aguilar, F.X., Cai, Z., D'Amato, A.W., 2014a. Non-industrial private forest owner's willingness-to-harvest: how higher timber prices influence woody biomass supply. *Biomass Bioenergy* 71, 202–215.
- Aguilar, F.X., Daniel, M.J., Cai, Z., 2014b. Family-forest owners' willingness to harvest Sawlogs and woody biomass: the effect of price on social availability. *Agric. Resour. Econ. Rev.* 43 (2), 279–299.
- Bailey, C., Dyer, J.F., Teeter, L., 2011. Assessing the rural development potential of lignocellulosic biofuels in Alabama. *Biomass Bioenergy* 35, 1408–1417.
- Bernard, H.R., 2011. *Research methods in anthropology: qualitative and quantitative approaches*, 5th ed. AltaMira Press, Walnut Creek, CA.
- Brandeis, C., Taylor, M., Abt, K.L., Alderman, D., Buehlmann, U., 2021. Status and trends for the u.s. forest products sector: a technical document supporting the forest service 2020 RPA assessment. U.S. Department of Agriculture Forest Service, Southern Research Station, Asheville, NC. Gen. Tech. Rep. SRS-258.
- Brown, R.C., Brown, T.R., 2012. Why are we producing biofuels? shifting to the ultimate source of energy. *Brownia*, Ames IA.
- Butler, B.J., Wear, D.N., 2013. Forest ownership dynamics of southern forests. In: Wear, D.N., Greis, J.G. (Eds.), *The southern forest futures project: technical report*. U.S. Department of Agriculture Forest Service, Southern Research Station, Asheville, NC, pp. 1–552. Gen. Tech. Rep. SRS-178.
- Butler, B.J., Butler, S.M., Floress, K., 2023. Studies of family forest owners in the USA: a systematic review of literature from 2000 through 2019. *Small-Scale Forest.* 22, 1–35.
- Butler, B.J., Sass, E.M., 2023. Wood supply from family forests of the United States: biophysical, social, and economic factors. *Forest Sci.* 69, 629–642.
- Carroll, W.D., Kapeluck, P.R., Harper, R.A., Van Lear, D.H., 2002. Background paper: historical overview of the southern forest landscape and associated resources. Wear, D.N. and Greis, J. G. *The Southern Forest Resource Assessment*. USDA Forest Service, Southern Research Station, Asheville, NC, pp. 583–605. Gen. Tech Rep. SRS-53.
- Cowdrey, A.E., 1996. *This Land, This South: An environmental history*. University Press of Kentucky, Lexington.
- D'Andrade, R., Strauss, C., 1992. *Human motives and cultural models*. Cambridge University Press, Cambridge.
- Dickens, E.D., Moorhead, D.J., Barger, C.T., Morris, L.A., Ogden, L.A., McElvany, B.C., 2012. A summary of pine straw yields and economic benefits in loblolly, longleaf and slash pine stands. *Agrofor. Syst.* 86, 315–321.
- Dwivedi, P., Alavalapati, J.R.R., 2009. Stakeholders' perceptions on forest biomass-based bioenergy development in the southern U.S. In: *Energy Policy*, 37, pp. 1999–2007.
- Fuentes, A., 2019. *Why we believe: evolution and the human way of being*. Yale University Press, New Haven.
- Gebru, B.M., Wang, S.W., Kim, S.J., Lee, W-K., 2019. Socio-ecological niche and factors affecting agroforestry practice adoption in different agroecologies of Southern Tigray, Ethiopia. *Sustainability*. 11, 3729.
- Hitchner, S., Schelhas, J., Brosius, J.P., 2016. Snake oil, silver buckshot, and people who hate us: metaphors and conventional discourses of wood-based bioenergy in the rural southeastern U.S. *Hum Organ* 73 (3), 204–217.
- Hitchner, S., Schelhas, J., Brosius, J.P., 2017. Even our Dairy Queen shut down": risk and resiliency in bioenergy development in forest-dependent communities in the U.S. *South. Econ. Anthropol.* 4, 186–199.
- Hitchner, S., Schelhas, J., Brosius, J.P., 2022. *Forests As Fuel: Energy, Landscape, Climate, and Race in the U.S. South*. Lexington Books, London.
- Hitchner, S., Dwivedi, P., Schelhas, J., Jagadish, A., 2019. Gatekeepers, shareholders, and evangelists: expanding communication networks of African American forest landowners in North Carolina. *Soc. Nat. Resour.* 32 (7), 751–767.
- Holland, D., Quinn, N., 1987. *Cultural models in language and thought*. Cambridge University Press, Cambridge.
- Joshi, O., Mehmood, S.R., 2011. Factors affecting nonindustrial private forest landowners' willingness to supply woody biomass for bioenergy. *Biomass Bioenergy* 35, 186–192.
- Lantinen, S.M., Song, N., Aguilar, F.X., 2014. Public policy promoting wood energy in the EU and US. In: Aguilar, F.X. (Ed.), *Wood energy in developed economies: resource management, economics, and policy*. Routledge, Oxon, pp. 189–222.
- Lejano, R., Ingram, M., Ingram, H., 2013. *The power of narrative in environmental networks*. MIT Press, Cambridge, MA.
- MacCleery, D.W., 2011. *American forests: a history of resiliency and recovery*. Forest History Society, Durham, NC.
- Mei, B., 2019. Timberland investments in the United States: a review and prospects. *For. Policy. Econ.* 109, 101998.
- Nelson, M.D., Ritters, K.H., Coulston, J.W., Domke, G.M., Greenfield, E.J., Langner, L.L., Nowak, D.J., O'Dea, C.B., Oswalt, S.N., Reeves, M.C., Wear, D.N., 2020. Defining the United States land base: a technical document supporting the USDA Forest Service 2020 RPA assessment. In: Gen. Tech. Rep. NRS-191. U.S. Department of Agriculture, Forest Service, Northern Research Station, Madison, WI.
- Newman, D.H., Wear, D.N., 1993. Production economics of private forestry: a comparison of industrial and nonindustrial forest owners. *Am J Agric Econ* 75 (3), 674–684.
- Ojiem, J.O., de Ridder, N., Vanlauwe, B., Giller, K.E., 2006. Socioecological niche: a conceptual framework for integration of legumes in smallholder farming systems. *Int. J. Agric. Sustain.* 4 (1), 79–93.
- Owen, W., 2002. The history of native plant communities in the South. In: Wear, D.N., Greis, J.G. (Eds.), *The southern forest resource assessment*. USDA Forest Service, Southern Research Station, Asheville, NC, pp. 47–61. Gen. Tech Rep. SRS-53.
- Paula, A.L., Bailey, C., Barlow, R.J., Morse, W., 2011. Landowner willingness to supply timber for biofuel: results of an Alabama survey of family forest owners. *Southern J. Appl. Forest.* 35 (2), 93–97.
- Perdue, J.H., Stanturf, J.A., Young, T.M., Huang, X., Dougherty, D., Pigott, M., Guo, Z., 2017. Profitability potential for *Pinus taeda* L. (loblolly pine) short-rotation bioenergy plantings in the southern USA. *For. Policy. Econ.* 83, 146–155.
- Quinn, N., 2005a. Introduction. Finding culture in talk: A collection of methods. Palgrave MacMillan, New York, pp. 1–34 edited by Naomi Quinn.
- Quinn, N., 2005b. How to reconstruct schemas people share, from what they say. In: Quinn, Naomi (Ed.), *Finding culture in talk: A collection of methods*. Palgrave MacMillan, New York, pp. 35–82.
- Rudel, T.K., 2023. *Reforesting the earth: the human drivers of forest conservation, restoration, and expansion*. Columbia University Press, New York.
- Sayer, J., Boedihartono, A.K., 2018. Integrated landscape approaches to forest restoration. In: Mansourian, S., Parrotta, J. (Eds.), *Forest landscape restoration: integrated approaches to support effective implementation*. Routledge, London, pp. 83–99.
- Schelhas, J., T.J. Brandeis, T.K. Rudel. 2021. Reforestation and natural regeneration in forest transitions: patterns and implications from the U.S. South. *Reg Environ. Change*. 21:8.
- Schelhas, J., Hitchner, S., Brosius, J.P., 2018. Envisioning and implementing wood-based bioenergy systems in the Southern United States: imaginaries in everyday talk. Special issue on "Energy and the Future. *Energy Res. Social Sci.* 35, 182–192.
- Schelhas, J., Pfeffer, M.J., 2005. Forest values of national park neighbors in Costa Rica. *Hum. Organ.* 64 (4), 385–397.
- Schelhas, J., Pfeffer, M.J., 2009. When global conservation meets local livelihoods: policy and Management Lessons. *Conserv. Lett.* 2 (6), 278–285.
- Stanturf, J.A., Vance, E.D., Fox, T.R., Kirst, M., 2013. Eucalyptus beyond its native range: environmental issues in exotic bioenergy plantations. *Internat. J. Forest. Res.* 2013, 463030.
- Straus, C., 1992. What makes Tony run? Schemas as motives reconsidered. in: *human motives and cultural models*. Cambridge University Press, Cambridge, pp. 191–224 edited by R. D'Andrade and C. Strauss.
- Strauss, C., 2005. Analyzing discourse for cultural complexity. Finding culture in talk: a collection of methods. Palgrave MacMillan, New York, pp. 204–242 edited by Naomi Quinn.
- Strauss, C., Quinn, N., 1997. *A Cognitive theory of cultural meaning*. Cambridge University Press, Cambridge.
- USDA Forest Service, 2021aa. Family forests (10+ acres) ownership characteristics: Southeastern United States, 2018. USDA Forest Service, Northern Research Station, Madison, WI. Res. Note NRS-261.
- USDA Forest Service, 2021ab. Family forests (10+ acres) ownership characteristics: South Central United States, 2018. USDA Forest Service, Northern Research Station, Madison, WI. Res. Note NRS-260.
- Williams, M., 1989. *Americans and their forests: a historical geography*. Cambridge University Press, Cambridge.